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Firearm with Double-Shot Trigger Assembly

Abstract

A firearm with a receiver body, a barrel, and a trigger assembly. The trigger assembly includes a trigger element movable between a pulled position and a released position. The trigger assembly is switchable between a safe mode that prevents the trigger element from moving to the pulled position, a semi-automatic mode that discharges a single round from the barrel responsive to each time the trigger element is moved to the pulled position, and a 2-shot mode that discharges an initial round from the barrel when the trigger element is moved to the pulled position and then a subsequent round when the trigger element is moved to the released position.

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Background/Summary

RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application Ser. No. 63/558,293 filed Feb. 27, 2024 for a “Firearm with Double-Shot Trigger Assembly,” which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates generally to firearms; in particular, this disclosure relates to a firearm with selectable firing modes to switch between semi-auto and two-shot firing.

BACKGROUND

[0003] There are many types of firearms, such as rifles and pistols, that are configured to fire rounds of ammunition. Firearms can have different firing modes, which dictates how many rounds are fired with a trigger pull. For semi-automatic firearms, a single round is fired each time the user pulls the trigger. Some firearms are fully-automatic, which means the firearm will fire continuously until the trigger is released or the magazine is empty. Another firing mode is burst, in which a limited number of shots are automatically fired each time the user pulls the trigger. In some cases, particularly for rifles, the firing mode is selectable between one or more modes.

SUMMARY

[0004] Accordingly to one aspect, this disclosure provides a firearm with a receiver body, a barrel, and a trigger assembly. The trigger assembly includes a trigger element movable between a pulled position and a released position. The trigger assembly is switchable between a safe mode that prevents the trigger element from moving to the pulled position, a semi-automatic mode that discharges a single round from the barrel responsive to each time the trigger element is moved to the pulled position, and a 2-shot mode that discharges a first round from the barrel when the trigger element is moved to the pulled position and then a subsequent round when the trigger element is moved to the released position.

[0005] According to another aspect, this disclosure provides a trigger assembly with a trigger element, a hammer, a sear, a safety selector, and a sear catch. The trigger element is movable between a pulled position and a released position. The hammer is movable between a cocked position and a fired position. The sear is configured to engage at least a portion of the hammer to retain the hammer in the ready position. In some cases, the sear is configured to release the hammer responsive to the trigger element moving to the pulled position. The safety selector is switchable between (i) a safe mode that prevents the trigger element from moving to the pulled position; (ii) a semi-automatic mode that discharges a single round responsive to each time the trigger element is moved to the pulled position; and (iii) a 2-shot mode that discharges a first round responsive to the trigger element moving to the pulled position and a subsequent round responsive to the trigger element moving to the released position. The sear catch is configured to prevent the sear from engaging the hammer in the 2-shot mode after discharging the first round responsive to moving the trigger element to the pulled position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The concepts described herein are illustrated by way of example and not by way of limitation in the accompanying figures. For simplicity and clarity of illustration, elements illustrated in the figures are not necessarily drawn to scale. Where considered appropriate, reference labels have been repeated among the figures to indicate corresponding or analogous elements.

[0007] FIG. 1 is a cross-sectional view of an example firearm along line 1-1 of FIG. 2;

[0008] FIG. 2 is a top view of an example firearm according to an embodiment of this disclosure;

[0009] FIG. 3 is a detailed left side view of the example firearm showing the safety selector in the safe position according to an embodiment of this disclosure;

[0010] FIG. 4 is a detailed left side view of the example firearm showing the safety selector in the semi-automatic position according to an embodiment of this disclosure;

[0011] FIG. 5 is a detailed left side view of the example firearm showing the safety selector in a two-shot burst position according to an embodiment of this disclosure;

[0012] FIG. 6 is a left side perspective view of an example trigger assembly according to an embodiment of this disclosure;

[0013] FIG. 7 is a left side perspective view of an example trigger element according to an embodiment of this disclosure;

[0014] FIG. 8 is a left side view of an example disconnect according to an embodiment of this disclosure;

[0015] FIG. 9 is a left side perspective view of an example sear catch according to an embodiment of this disclosure;

[0016] FIG. 10 is a left side perspective view of an example trigger housing according to an embodiment of this disclosure;

[0017] FIG. 11 is a left side view of an example bowl according to an embodiment of this disclosure; and

[0018] FIG. 12 is a left side perspective view of an example sear according to an embodiment of this disclosure;

[0019] FIG. 13 is a left side view of an example hammer according to an embodiment of this disclosure;

[0020] FIG. 14 is a left side perspective view of an example safety selector according to an embodiment of this disclosure;

[0021] FIG. 15 is a left side view of an example safety pen according to an embodiment of this disclosure;

[0022] FIG. 16 is a left side view of an example bowl catch according to an embodiment of this disclosure;

[0023] FIG. 17 is a left side cross-sectional view of the example trigger assembly with the safety selector in the safe position according to an embodiment of this disclosure;

[0024] FIG. 18 is a left side cross-sectional view of the example trigger assembly with the safety selector in the semi-auto position and the trigger pulled according to an embodiment of this disclosure;

[0025] FIG. 19 is a left side cross-sectional view of the example trigger assembly with the safety selector in the semi-auto position and the trigger pulled after the hammer has been released to impact the firing pin according to an embodiment of this disclosure;

[0026] FIG. 20 is a left side cross-sectional view of the example trigger assembly with the safety selector in the semi-auto position after discharging a round while the trigger is still pulled and the hammer has re-cocked and been retained by the disconnect according to an embodiment of this disclosure;

[0027] FIG. 21 is a left side cross-sectional view of the example trigger assembly with the safety selector in the semi-auto position after discharge when the trigger is released to the reset position in which the hammer is retained by the sear according to an embodiment of this disclosure;

[0028] FIG. 22 is a left side cross-sectional view of the example trigger assembly with the safety selector in the two-shot position and the trigger pulled according to an embodiment of this disclosure;

[0029] FIG. 23 is a left side detailed perspective view showing the sear hooked onto the sear latch after pulling the trigger according to an embodiment of this disclosure;

[0030] FIG. 24 is a left side cross-sectional view of the example trigger assembly in the two-shot

position after pulling the trigger to release the hammer and impact the firing pin according to an embodiment of this disclosure;

[0031] FIG. **25** is a detailed left side view of the example trigger assembly in the two-shot position after the hammer has re-cocked and caught on the disconnect according to an embodiment of this disclosure;

[0032] FIG. **26** is a detailed perspective view showing the sear being held down by the sear catch in the two-shot mode to allow the hammer to be released in fire another round according to an embodiment of this disclosure;

[0033] FIG. **27** is a detailed left side view showing a pin on the hammer contacting the sear catch to disengage the sear after the hammer has released according to an embodiment of this disclosure;

[0034] FIG. **28** is a detailed left side view showing the sear retaining the hammer after firing the second round according to an embodiment of this disclosure;

[0035] FIG. **29** is a side cross-sectional view of the example trigger assembly with the bolt locked in a back position after firing the last round in the magazine according to an embodiment of this disclosure;

[0036] FIGS. **30** and **31** is a detailed with side view of the trigger assembly showing a safety pin pushing the sear catch forward to release the sear when the bolt is locked back when the trigger is pulled and released, respectively, according to an embodiment of this disclosure;

[0037] FIG. **32** is a detailed side view of the trigger assembly after the bolt is released to show the safety pin preventing the hammer from following the bolt forward according to an embodiment of this disclosure;

[0038] FIG. **33** is a detailed cross-sectional view of the trigger assembly in the semi-auto mode with the trigger pulled to show that the pocket in the safety selector is shallow enough to prevent the sear catch from hooking the sear;

[0039] FIG. **34** is a detailed cross-sectional view of the trigger assembly in the semi-auto mode showing that with the trigger in the pulled position that the groove in the safety selector acts as a secondary safety to prevent the sear from getting hooked by the sear catch;

[0040] FIG. **35** is a detailed cross-sectional view of the trigger assembly in the safe mode showing that the safety selector prevents pulling of the trigger according to an embodiment of this disclosure; and

[0041] FIG. **36** is a detailed cross-sectional view of the trigger assembly showing that while the sear is not fixed to the trigger, the groove in the safety selector prevents the sear from rotating enough to release the hammer according to an embodiment of this disclosure.

DETAILED DESCRIPTION OF THE DRAWINGS

[0042] While the concepts of the present disclosure are susceptible to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the drawings and will be described herein in detail. It should be understood, however, that there is no intent to limit the concepts of the present disclosure to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives consistent with the present disclosure and the appended claims.

[0043] References in the specification to “one embodiment,” “an embodiment,” “an illustrative embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may or may not necessarily include that particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. Additionally, it should be appreciated that items included in a list in the form of “at least one A, B, and C” can mean (A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C). Similarly, items listed in the form of “at least one of A, B, or C” can mean

(A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C).

[0044] In the drawings, some structural or method features may be shown in specific arrangements and/or orderings. However, it should be appreciated that such specific arrangements and/or orderings may not be required. Rather, in some embodiments, such features may be arranged in a different manner and/or order than shown in the illustrative figures. Additionally, the inclusion of a structural or method feature in a particular figure is not meant to imply that such feature is required in all embodiments and, in some embodiments, may not be included or may be combined with other features.

[0045] FIGS. 1 and 2 illustrate an example firearm 10 according to an embodiment of this disclosure. The firearm 10 is shown for purposes of example, but some of the components shown could be optional and additional components could be provided in some circumstances. Although the firearm 10 is shown as an AR-15 variant for purposes of example, any firearm configuration, such as handguns, rifles, shotguns, personal defense weapons, etc., are within the scope of this disclosure. As shown, the firearm 10 includes a receiver body 12, a barrel 14, a grip 16, a buffer tube 18, and trigger assembly 20 with a trigger 22. In the example shown, the firearm 10 includes a magazine well 24 to receive a magazine (not shown) to feed rounds of ammunition to be discharged.

[0046] The receiver body 12 is configured to house the trigger assembly 20 and associated components as found therein. A bolt assembly (not shown) can be slidably disposed in the receiver body 12 for axially reciprocating the recoil movement during a firing sequence of the firearm 10. The bolt assembly is configured to interface with the trigger assembly 20 as described herein.

[0047] The barrel 14 is positioned at the forward in of the receiver body 12. The barrel 14 provides a path to release rounds of ammunition therethrough as the firearm 10 discharges. The grip 16 provides a point of support for the shooter of the firearm 10 and can be held by the shooter's hand, including when operating the trigger assembly 20. The grip 16 assists the shooter in stabilizing the firearm 10 during firing and manipulation of the firearm 10. In some embodiments, the grip 16 is mounted to the receiver body 12. At the rearward portion of the receiver body is a buffer tube 18 that receives a portion of the bolt assembly (not shown) during operation and to which a stock (not shown) can be attached. The magazine well 24 is configured to receive a magazine (not shown) for feeding ammunition within the firearm 10. The trigger assembly 20 includes a trigger 22 configured to be pulled by the finger of the shooter (e.g., the index finger) to initiate the firing sequence of the firearm 10.

[0048] Referring to FIGS. 3-5, the firearm 10 includes a safety selector 26 that is movable to select different firing modes. As shown, the safety selector 26 pivots between a safe mode (FIG. 3), a semi-automatic mode (FIG. 4), and a two-shot mode (FIG. 5). In the example shown, the safety selector 26 includes a lever 28 that can be manipulated by the user to move the safety selector 26 between firing modes and a pointer 30 that indicates which firing mode is selected. As shown, the surface of the receiver body 12 includes catches 29 to retain the lever 28 in the selected firing mode. In some embodiments, the receiver body 12 may include surface ornamentation, embossing, and/or labels to specify which position corresponds to which firing mode.

[0049] Referring now also to FIG. 14, the safety selector 26 includes a cylindrical portion 32 with a first pocket 34 into second pocket 36. As explained herein, in the safe mode, the cylindrical portion 32 blocks a pin 23 of the trigger element 40 (see FIG. 7) from being pulled. The first pocket 34 aligns with the pin 23 of the trigger element 40 in the semi-automatic mode. The second pocket 36 aligns with the pin 23 of the trigger 22 in the two-shot mode. In the embodiment shown, the cylindrical portion 32 includes a groove 38 that slidably receives a leg 44 of the sear 42 to function as a secondary safety as discussed herein.

[0050] Referring now to FIG. 6, there is shown a trigger assembly 20 according to an embodiment of this disclosure. In the embodiment shown, the trigger assembly 20 includes a trigger housing 46 with a flange 48 that extends around the periphery and has an open end. As shown, the trigger

housing **46** holds other components of the trigger assembly **20**. The flange **48** defines openings through which a first pin **50** and a second pin **52** extend. In the embodiment shown, the trigger assembly **20** includes a trigger element **40** with a trigger **22** that pivots about the second pin **52**. The trigger **22** is urged towards the rest position with a biasing member, such as a spring **62**. As explained herein, pivoting of the trigger element **40** causes a sear engagement portion **53** to actuate the sear **42**, which releases the hammer **54** to impact the firing pin (not shown) and discharge a round from the barrel **14**. The hammer **54** pivots about a third pin **60** and is urged towards the firing pin with a biasing member, such as a spring **64**. Due to recoil of a bolt **56** (FIG. **11**), the hammer **54** will be re-cocked and be hooked by a disconnect **58** until the trigger **22** is reset and then the sear **42** will retain the hammer **54** for the next shot depending on which firing mode is selected.

[0051] Referring to FIG. **7**, there is shown in example trigger element **40**. In the embodiment shown, a forward portion includes a sear engagement portion **53** that is movable when the shooter pulls the trigger **22** to engage the sear **42**, which releases the hammer **54**. There are openings **66** dimensioned to receive the first pin **50** and about which the trigger element **40** pivots. At the rearward portion, there is a pin **23** that it extends from the trigger element **40**. As explained herein, in the safe position of the safety selector **26**, the cylindrical portion **32** abuts the pin **23**, which prevents the trigger **22**. In the semi-automatic and two-shot firing modes, the pin **23** is arranged to be received by the first pocket **34** and the second pocket **36**, respectively. The depth of the first pocket **34** compared to the depth of the second pocket **36** determines the amount of the pin **23** that can be received, and thereby the movement of the trigger element **40**.

[0052] FIG. **8** illustrates an embodiment of the disconnect **58**. In the embodiment shown, there is a hook portion configured to retain the hammer **54** upon recocking. As shown, there is an opening **68** configured to receive the first pin **50** and pivot about the first pin **50**.

[0053] FIG. **9** shows an example of a sear catch **70** with an opening **71** through which the second pin **52** is received. The sear catch **70** is configured to pivot such the a ledge **72** holds down the sear **42** in the 2-shot firing mode to allow a second shot. As shown, the sear catch **70** includes an extension **74** that, as explained below, is actuated by the hammer **54** to disengage the sear catch **70** from the sear **42** after the first shot in the 2-shot firing mode, as explained herein.

[0054] FIG. **10** shows an illustrative embodiment of the trigger housing **46** to house components of the trigger assembly **20**. As shown, the flange **48** extends around the periphery to define a first set of openings **76** that receives the first pin **50** and a second set of openings **78** that receives the second pin **52**.

[0055] FIG. **12** shows an example sear **42** that retains the hammer **54** back in a cocked position until the shooter pulls the trigger **22**, and then releases the hammer **54** to discharge a round in the semi-automatic firing mode. As shown, the sear **42** includes a forward end with a retention ledge **80** that is configured to retain the hammer **54** in a cocked position. There is a first segment **82** and a second segment **84** extending from the retention ledge **80**. There is a first opening **86** in the first segment **82** and a second opening **88** in the second segment **84** to receive the first pin **50** therebetween. The sear **42** is configured to pivot about the first pin **50**. A leg **44** extends from the rearward end and is dimensioned to be received in the groove **38** in the safety selector **26**.

[0056] Referring now to FIG. **13**, there is shown an example hammer **54** with an opening **89** dimensioned to receive the third pin **60** upon which the hammer **54** rotates. The hammer **54** includes a firing engagement surface **90** configured to impact the firing pin when the hammer **54** is released to discharge a round from the firearm **10**. The hammer **54** includes a hook portion **92** configured to engage with the disconnect **58** upon re-cocking. There is a notch **94** configured to be engaged the sear **42** to selectively control when the hammer **54** is released to discharge a round.

[0057] FIG. **14** shows an example safety selector **26** with the lever **28** that can be manipulated (e.g., rotated) by the shooter to select a desired firing mode, which is indicated by a pointer **30**. As discussed herein, the safety selector **26** includes a cylindrical portion **32** with a first pocket **34** and a second pocket **36** into which the pin **23** of the trigger element **40**. In some embodiments, the depth

by which the pin **23** is received within the pockets **34**, **36** impacts the amount of travel of the trigger element **40**, which determines whether the firearm **10** is in the semi-automatic or 2-shot mode firing mode. As discussed herein, the groove **38** is configured to receive the leg **44** of the sear **42** to act as a secondary safety. FIG. **15** shows an example safety pin **96** with a head **98**. FIG. **16** shows an example bolt catch **100** that is configured to retain the bolt **56** in the back position upon discharging the last round in the magazine (not shown).

[0058] FIG. **17** shows the trigger assembly **20** with the safety selector **26** in the safe mode. In the safe mode, the safety selector **26** is rotated to a position in which neither of the pockets **34**, **36** are aligned with the pin **23** of the trigger element **40**. Instead, the cylindrical portion **32** of the safety selector **26** abuts the pin **23** to block movement of the trigger element **40**, and therefore preventing the trigger **22** from being pulled by the shooter. Since the trigger **22** cannot move, the trigger element **40** cannot push down the sear **42** to release the hammer **54**.

[0059] FIGS. **18-21** are progressive views of the example trigger assembly in which the safety selector **26** is in the semi-automatic mode after the trigger **22** has been pulled and a round discharged through re-cocking of the hammer **54**. As can be seen the safety selector **26** is configured such that the first pocket **34** is aligned with the pin **23** of the trigger element **40**. This allows the trigger **22** to travel a limited distance based on the depth of the first pocket **34**. In some embodiments, the first pocket **34** is more shallow than the second pocket **36**. The depth of the pocket **34** is configured to allow the trigger element **40** to travel just enough to move the sear enough to allow the hammer to release. As shown in FIG. **18**, the trigger **22** has travelled enough for the pin **23** to be received within the first pocket **34**. This amount of travel allows the sear engagement portion **53** to engage and push down the sear **42**, which releases the hammer **54**.

[0060] FIG. **18** shows the initial release of the hammer **54** after the trigger **22** pull. FIG. **19** shows the hammer **54** continuing forward movement (counter-clockwise in this view) to impact the firing pin **102** to discharge a round **104** out of the barrel **14**. FIG. **20** shows the hammer **54** being re-cocked by the bolt **56** (rotating clockwise in this view). This re-cocking movement causes the hook portion **92** of the hammer **54** to catch on the hook portion **66** of the disconnect **58**. As seen, there is a biasing member **106** between the disconnect **58** and the trigger element **40**. This spring-loads the disconnect **58** against the trigger element **40** so the disconnect **58** can latch onto the hammer **54** while the trigger is in the pulled position. Referring to FIG. **21**, as the trigger **22** is released, the hammer **54** will release from the disconnect **58**. This will cause the notch **94** of the hammer **54** to catch on the sear **42**. This is commonly referred to as resetting the trigger. The sear **42** and the trigger **22** are spring-loaded against the trigger housing **46** to urge the sear **42** and the trigger **22** to the reset position. The sear **42** will retain the hammer **54** until the next time the shooter pulls the trigger **22**.

[0061] FIGS. **22-28** are progressive views of the example trigger assembly in which the safety selector **26** is in the 2-shot mode after the trigger **22** has been pulled and a first round discharged through re-cocking of the hammer **54** to discharge another shot, and re-cocked again to the reset position. As can be seen, the safety selector **26** is configured such that the second pocket **36** is aligned with the pin **23** of the trigger element **40** in the 2-shot mode. This allows the trigger **22** to travel a larger distance based on the deeper depth of the second pocket **36** compared to the first pocket **34**. The depth of the second pocket **36** is configured to allow the trigger element **40** to travel a larger range of motion. This additional travel of the trigger element moves the sear **42** an additional distance to not only allow the hammer **56** to release (FIG. **22**), but continue to travel (rotate counter-clockwise in this example) enough for the sear **42** to hook onto the sear catch **70** (FIG. **23**). In the embodiment shown, the retention ledge **80** of the sear catch **70** includes a surface **106** that creates an interference connection with the sear **42** that retains the sear **42** in a lowered position that prevents the sear **42** from hooking to the hammer **56**. This means when the trigger element **40** moves to the reset position when the trigger **22** is released by the shooter, the sear **42** will not move to a position to retain the hammer due to being hooked by the sear catch **70**, as

discussed herein.

[0062] As shown in FIG. 24, the hammer 56 continues forward to impact the firing pin 102 to discharge a first round 104. FIG. 25 shows the hammer 54 being re-cocked by the bolt 56 (rotating clockwise in this view) to cause the hook portion 92 of the hammer 54 to catch on the hook portion 66 of the disconnect 58, like the semi-automatic mode. The hammer 54 will remain hooked to the disconnect 58 until the shooter releases the trigger 22. When the trigger 22 is released, the hammer 54 will no longer be retained by the disconnect 58. Instead of the trigger 22 resetting and the hammer 54 being retained by the sear 42 in the ready to fire position when released like in the semi-automatic mode, however, the hammer 54 will be released and fire another round because the sear 42 is being held down and out of the way of the hammer 54 by the sear catch 70 as shown in FIG. 26. This allows the hammer 54 to be released again to discharge a second round. Now referring to FIG. 27, when the hammer 54 releases (rotates counter-clockwise) to fire the second round, the extension 74 of the sear catch 70 is engaged by a pin 108 on the hammer 54 to release the sear 42 from the sear catch 70. The sear 42 then is able to move into a position to catch onto the hammer 54 after the second round is fired. After the second round is fired, as shown in FIG. 28, the hammer 54 will be pushed back by the bolt 56 and can now be caught by the sear 42 to retain the hammer 54 in the ready position. The shooter can then pull and release the trigger 22 again to fire two additional rounds.

[0063] Referring now to FIGS. 29-32, operation of the safety pin 96 is shown in the 2-shot firing mode. The safety pin 96 solves a potential safety issue that could occur when the safety selector 26 is in the 2-shot mode and the bolt 56 is locked back. When the bolt 56 is locked back with the bolt catch 100, which could occur when the magazine (not shown) is out of rounds, the sear 50 could be held down by the sear catch 70 and will not catch the hammer 54 so it is retained in the ready to fire position. This could happen, for example, if the initial round is fired when the magazine runs out of ammo, and the second round is not able to be fired. If the shooter then inserts a loaded magazine into the firearm 10 and releases the bolt 56, the hammer 54 may follow the bolt 56 when it is released, and inadvertently discharge a round, without the safety pin 96. As explained herein, the safety pin 96 disengages the sear catch 70 from the sear 50 to prevent the hammer 54 from following the bolt 56 forward and inadvertently discharging a round when the bolt 56 is released. As shown in FIG. 29, in the embodiment shown, the bolt 56 is locked back to the rear by the bolt catch 100 once the last round is fired. As shown, the safety pin 96 has a first end against a surface of the trigger 22 and a second end against a surface of the sear catch 70. This means when the trigger 22 moves to a released position, the trigger's surface will act upon the safety pin 96 to correspondingly move the other end of the safety pin 96 against the sear catch 70 to disengage the sear catch 70 from the sear 56. FIG. 30 shows engagement of the sear 42 by the sear catch 70 when the trigger 22 is pulled in the 2-shot firing mode. Since there are no additional rounds, when the trigger 22 is released, the safety pin 96 will push the sear catch 70 forward to release the sear 42. Referring now to FIG. 31, since the safety pin 96 disengaged the sear catch 70 from the sear 42, the sear 42 will catch the notch 94 on the hammer 54. This prevents the hammer 54 from following the bolt forward should the bolt be released to the forward position as shown by FIG. 32.

[0064] FIGS. 33-36 illustrate operation of a secondary safety in which the leg 44 of the sear 40 rides in the groove 38 (see also FIGS. 6, 12 and 14) of the safety selector 26 in the embodiment shown. As discussed herein, the length of travel of the sear 42 is based on the orientation of the safety selector 26. As discussed, the pockets 34, 36 have different depths that result in different travel distances of the trigger element 40 when the trigger 22 is pulled, thereby changing the travel length of the sear 42. Referring to FIG. 33, which shows the semi-automatic firing mode, the trigger 22 cannot rotate far enough to have the sear get hooked on the sear catch 70 because the pin 23 extending from the trigger element 40 bottoms out in the pocket 34 of the safety selector 26. As shown in FIGS. 34-36, which shows the safety selector 26 in the semi-automatic mode, the groove 38 of the safety selector 26 acts as a secondary safety from the sear 42 getting hooked by the sear

catch 70. If the firearm 10 has strong recoil or there is excess debris in the system, the trigger 22 could begin functioning like it is in the 2-shot mode while in the semi-automatic mode. However, the secondary safety prevents that from occurring. In the safe mode (FIG. 35), the trigger 22 cannot be pulled because the pin 23 is against the cylindrical portion 32 of the safety selector 26. Although the trigger 22 cannot be pulled, the sear 42 is not fixed to the trigger 22 or trigger element 40. However, if the sear 42 were to be manipulated while the safety selector is in the safe position, the leg 44 of the sear 42 would contact the safety selector 26 which would not allow the sear 42 to rotate far enough to release the hammer 54.

Claims

1. A firearm comprising: a receiver body; a barrel extending from the receiver housing; a trigger assembly with a trigger element movable between a pulled position and a released position, wherein the trigger assembly is switchable between (i) a safe mode that prevents the trigger element from moving to the pulled position; (ii) a semi-automatic mode that discharges a single round from the barrel responsive to each time the trigger element is moved to the pulled position; and (iii) a 2-shot mode that discharges an initial round from the barrel when the trigger element is moved to the pulled position and then a subsequent round when the trigger element is moved to the released position.
2. The firearm of claim 1, wherein the trigger assembly further comprises: a hammer movable between a cocked position and a fired position; a sear configured to engage at least a portion of the hammer to retain the hammer in the ready position, wherein the sear is configured to release the hammer responsive to the trigger element moving to the pulled position; a sear catch configured to prevent the sear from engaging the hammer in the 2-shot mode after discharging the initial round from the barrel.
3. The firearm of claim 2, wherein the hammer is configured to disengage the sear catch from the sear responsive to moving towards the fired position subsequent to discharging the initial round.
4. The firearm of claim 3, wherein the hammer includes a transversely extending pin that is configured to disengage the sear catch from the sear responsive to the hammer moving towards the fired position subsequent to discharging the initial round.
5. The firearm of claim 2, further comprising a safety selector configured to select between the safe mode, the semi-automatic mode, and the 2-shot mode.
6. The firearm of claim 5, wherein the safety selector includes a cylindrical portion that blocks the trigger element from moving to the pulled position in the safe mode.
7. The firearm of claim 6, wherein the cylindrical portion includes a first pocket having a first depth and a second pocket having a second depth that is deeper than the first depth.
8. The firearm of claim 7, further comprising a pin extending from the trigger element, wherein the pin is received in the first pocket when the safety selector is in the semi-automatic mode.
9. The firearm of claim 8, wherein the pin is received in the second pocket in the 2-shot mode.
10. The firearm of claim 9, wherein the depth of the first pocket is dimensioned to limit travel of the trigger element to actuate the sear to release the hammer, but not allow the sear to be engaged by the sear catch.
11. The firearm of claim 10, wherein the depth of the second pocket is dimensioned to allow the trigger element to travel a distance that both actuates the sear to release the hammer and allows the sear to be engaged by the sear catch.
12. The firearm of claim 6, further comprising a secondary safety that limits travel of the sear in the semi-automatic mode to prevent the sear catch from engaging the sear.
13. The firearm of claim 12, wherein the secondary safety comprises a groove in the cylindrical portion of the safety selector in which a leg of the sear rides, wherein in the semi-automatic mode, the groove limits travel of the sear to prevent engagement of the sear by the sear catch.

- 14.** The firearm of claim 6, further comprising a safety pin configured to disengage the sear catch from the sear responsive to a last round being discharged.
- 15.** A trigger assembly comprising: a trigger element movable between a pulled position and a released position; a hammer movable between a cocked position and a fired position; a sear, independently movable with respect to the trigger element, configured to engage at least a portion of the hammer to retain the hammer in the ready position, wherein the sear is configured to release the hammer responsive to the trigger element moving to the pulled position; a safety selector switchable between (i) a safe mode that prevents the trigger element from moving to the pulled position; (ii) a semi-automatic mode that discharges a single round responsive to each time the trigger element is moved to the pulled position; and (iii) a 2-shot mode that discharges an initial round responsive to the trigger element moving to the pulled position and a subsequent round responsive to the trigger element moving to the released position; and a sear catch configured to prevent the sear from engaging the hammer in the 2-shot mode after discharging the initial round responsive to moving the trigger element to the pulled position.
- 16.** The trigger assembly of claim 15, wherein the hammer includes a transversely extending pin that is configured to disengage the sear catch from the sear responsive to the hammer moving towards the fired position subsequent to discharging the initial round.
- 17.** The trigger assembly of claim 15, wherein the safety selector includes a cylindrical portion that blocks the trigger element from moving to the pulled position in the safe mode.
- 18.** The trigger assembly of claim 17, wherein the cylindrical portion includes a first pocket having a first depth and a second pocket having a second depth that is deeper than the first depth.
- 19.** The trigger assembly of claim 18, further comprising a pin extending from the trigger element, wherein the pin is configured to be received in the first pocket when the safety selector is in the semi-automatic mode.
- 20.** The trigger assembly of claim 19, wherein the pin is configured to be received in the second pocket in the 2-shot mode.
- 21.** The trigger assembly of claim 20, wherein the depth of the first pocket is dimensioned to limit travel of the trigger element to actuate the sear to release the hammer when the trigger element moves to the pulled position, but not allow the sear to be engaged by the sear catch.
- 22.** The trigger assembly of claim 21, wherein the depth of the second pocket is dimensioned to allow the trigger element to travel a distance that actuates both the sear to release the hammer and allows the sear to be engaged by the sear catch.
- 23.** The trigger assembly of claim 15, further comprising a secondary safety that limits travel of the sear in the semi-automatic mode to prevent the sear catch from engaging the sear.
- 24.** The trigger assembly of claim 23, wherein the secondary safety comprises a groove in the cylindrical portion of the safety selector in which a leg of the sear rides, wherein in the semi-automatic mode, the groove limits travel of the sear to prevent engagement of the sear by the sear catch.
- 25.** The trigger assembly of claim 15, further comprising a safety pin configured to disengage the sear catch from the sear responsive to a last round being discharged.
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